

Applied Labor Economics

Shortage Lists and Monopsony

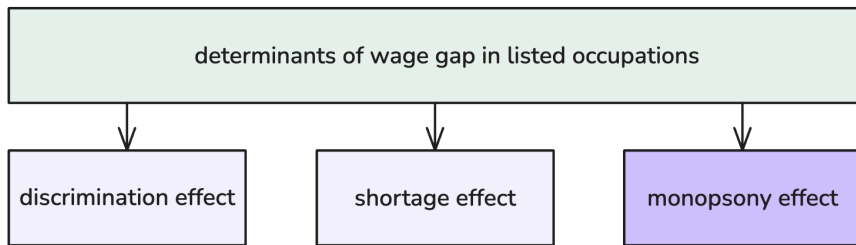
Project Update

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Context - *quick recap*

Question: *what is the causal effect on wages of being a non-EEA worker in a listed occupation ?*



How to answer: DiD comparing natives, eea and non-eea, across listed and non-listed occupations, before and after the policy. For culturally matched samples.

Context - *so far*

What we have done so far:

1. A lot of data cleaning.
2. Some descriptives.
3. Some regressions (*on the 2021 reform*).
4. Some update on our thinking.

1. data cleaning

2. descriptives

3. regressions

4. thinking

Setting - *evidence on métiers en tension*

Provence-Alpes-Côte d'Azur	0.55	0.57	0.45	0.51	0.36	0.62	0.56	0.54	0.54	0.51	0.57	0.49	1
Pays de la Loire	0.47	0.64	0.51	0.59	0.38	0.5	0.51	0.43	0.51	0.49	0.46	1	0.49
Occitanie	0.54	0.51	0.43	0.51	0.36	0.56	0.49	0.41	0.48	0.54	1	0.46	0.57
Nouvelle-Aquitaine	0.57	0.54	0.51	0.56	0.3	0.5	0.45	0.42	0.57	1	0.54	0.49	0.51
Normandie	0.51	0.62	0.53	0.62	0.3	0.46	0.59	0.49	1	0.57	0.48	0.51	0.54
Ile-de-France	0.47	0.49	0.41	0.44	0.27	0.38	0.5	1	0.49	0.42	0.41	0.43	0.54
Hauts-de-France	0.47	0.51	0.41	0.51	0.37	0.51	1	0.5	0.59	0.45	0.49	0.51	0.56
Grand Est	0.59	0.58	0.41	0.51	0.44	1	0.51	0.38	0.46	0.5	0.56	0.5	0.62
Corse	0.38	0.38	0.35	0.36	1	0.44	0.37	0.27	0.3	0.3	0.36	0.38	0.36
Centre-Val de Loire	0.58	0.68	0.5	1	0.36	0.51	0.51	0.44	0.62	0.56	0.51	0.59	0.51
Bretagne	0.47	0.5	1	0.5	0.35	0.41	0.41	0.41	0.53	0.51	0.43	0.51	0.45
Bourgogne-Franche-Comté	0.56	1	0.5	0.68	0.38	0.58	0.51	0.49	0.62	0.54	0.51	0.64	0.57
Auvergne-Rhône-Alpes	1	0.56	0.47	0.58	0.38	0.59	0.47	0.47	0.51	0.57	0.54	0.47	0.55

region	2021	2024	2025
Auvergne	30	2	29
PACA	26	4	30
N. Aquitaine	30	2	26
Bourgogne	30	4	22
Centre	30	3	20
Occitanie	30	2	21
PdL	30	4	18
HdF	30	2	19
IdF	30	4	17
Grand Est	21	4	25
Normandie	30	4	15
Bretagne	30	3	13
Corse	14	4	20

Setting- *balance checks*

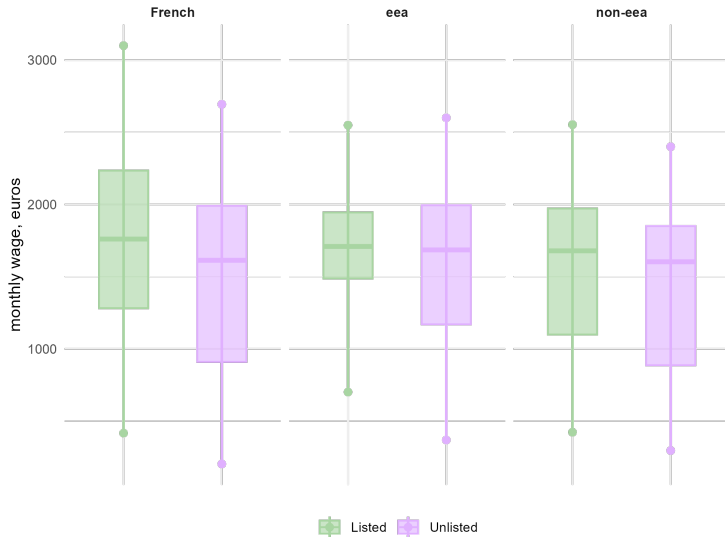
Takeaway: Variability on observables (easy to control). Sample differences (less easy).

	<i>Treatment Status</i>		<i>Permit Status</i>		
	<i>Unlisted</i>	<i>Listed</i>	<i>French</i>	<i>EEA</i>	<i>Non-EEA</i>
Age (years)	31.63	33.19	32.60	36.54	38.45
Monthly wage	1,588	1,771	1,720	1,653	1,560
Tenure (months)	8.00	7.61	7.76	5.90	6.63
Diploma (%)	66	66	67	33	35
Male (%)	49	61	57	64	66
Experience (years)	3.91	4.49	4.30	4.08	5.59
Obs. (people)	5,904,350	2,745,382	6,540,534	50,095	77,394
Obs. (contracts)	18,266,634	43,513,387	60,214,080	681,184	88,4757

NB: data 2019-2025.

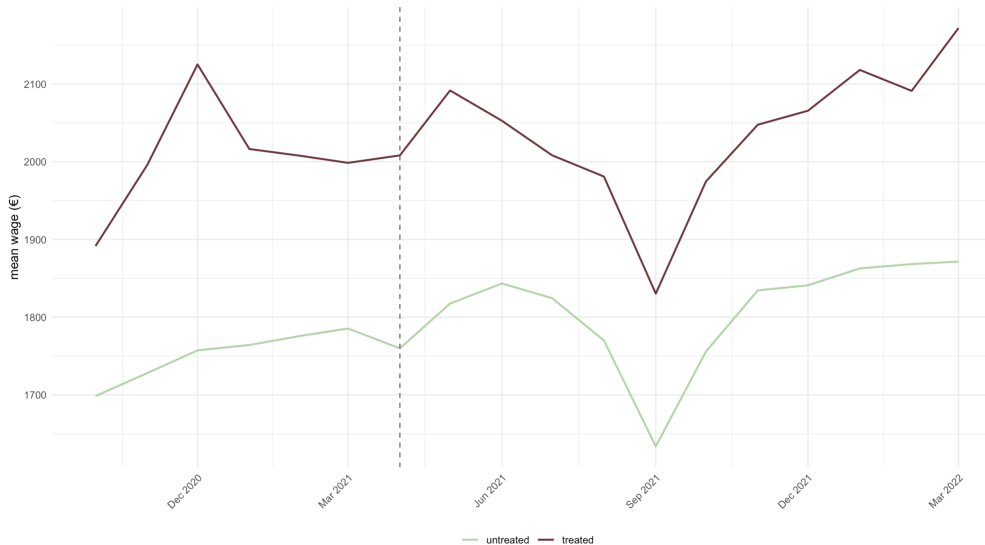
Setting- *evidence on wage gap*

Takeaway: There is a wage gap, and the shortage effect is real.



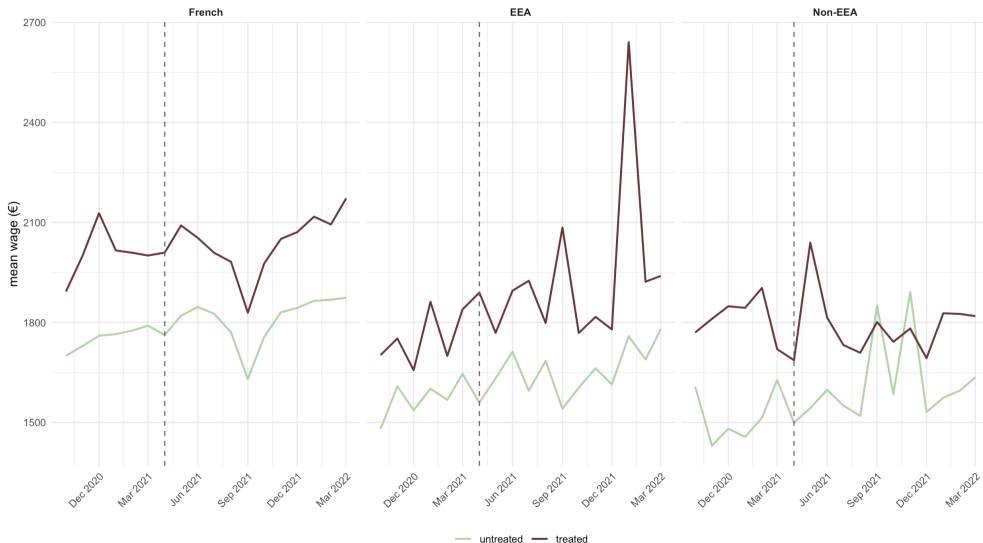
Results - *pre-trends*

Takeaway: Pre-trends for the 2021 reform do not seem to hold.



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Results - *specification 1*

Spec. 1: DDD

Variation: comparing wage changes for different types of workers. Across fap*region that became listed after April 2021 vs. those that did not.

$$\begin{aligned}\ln(w_{ijrt}) = & \alpha + \beta_1 \text{EEA}_i + \beta_2 \text{NonEEA}_i + \beta_3 \text{post}_t + \mathbf{X}'_{ijrt} \boldsymbol{\gamma} \\ & + \beta_4 (\text{French}_i \times \text{treatment}_{jr} \times \text{post}_t) + \beta_5 (\text{EEA}_i \times \text{treatment}_{jr} \times \text{post}_t) \\ & + \beta_6 (\text{NonEEA}_i \times \text{treatment}_{jr} \times \text{post}_t) + \delta_{jr} + \delta_t + \varepsilon_{ijrt}\end{aligned}$$

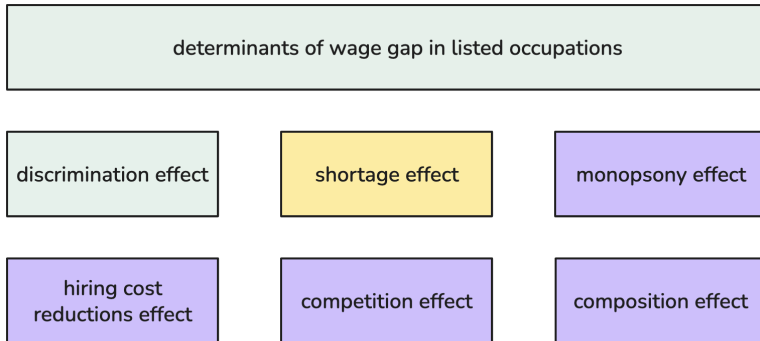
- ▶ β_1, β_2 : baseline wage gap, unconditional on listing.
- ▶ β_3 : common wage trend after April 2021.
- ▶ β_4 : wage change for French workers in listed after Apr2021.
- ▶ β_5 : differential wage change for EEA workers in listed after Apr2021 (*should be 0*).
- ▶ β_6 : differential wage change for non-EEA workers in listed cells after Apr2021 (*should be negative*).

Results - *specification 1*

<i>(ref: french)</i>	<i>estimate</i>	<i>std. error</i>	<i>p value</i>
EEA	0.0625	0.0119	<0.001 ***
Non-EEA	-0.0464	0.0108	<0.001 ***
Post	0.0330	0.0038	<0.001 ***
Male	0.1262	0.0153	<0.001 ***
Diploma	0.0379	0.0031	<0.001 ***
Experience (months)	0.0008	0.0001	<0.001 ***
French $\times$ treatment $\times$ post	-0.0255	0.0082	0.002 **
EEA $\times$ treatment $\times$ post	-0.0045	0.0220	0.837
Non-EEA $\times$ treatment $\times$ post	-0.0412	0.0208	0.048 *

Notes: 14,147,973 observations. Fixed effects: occupation $\times$ region (2,629), year (6). Standard errors clustered at occupation $\times$ region level.

Conclusions - *what are we actually capturing ?*



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<i>effect</i>	<i>mechanism</i>	<i>direction</i>	<i>consequence</i>
Monopsony	permit collapses non-EEA outside option, EEA workers unaffected	↓	target parameter
Competition	more non-EEA workers enter listed occupations post-listing, EEA inflow unaffected	↓	confounded
Hiring cost	labour market test waived only for non-EEA workers, employer may capture surplus	ambiguous	confounded
Composition	listing attracts a different type of non-EEA worker post-treatment	↓	confounded
Shortage	labour scarcity raises wages for all workers	↑	differenced out.
Discrimination	taste-based penalties depress non-EEA wages	↓	if time-invariant differenced out. if time-varying, managed by samples.

Conclusions - *what's next*

Recent data

- ▶ Use 2025S2 data to study May 2025 effects, *anybody managed to open it in SAS?*

Identification

- ▶ Staggered DiD + other simpler regressions and event studies.
- ▶ Incorporate better *niveau de formation* measures.

Robustness

- ▶ Did our sample composition change ?
- ▶ Where are our workers located in France ?
- ▶ All of our sample was registered in France Travail. How does it affect the generality of our results ?

Thank you for listening!

Any questions or feedback ?

Appendix - *data cleaning procedure*

Ficher Historique → *socio-demographic data*.

- ▶ Filter: relevant nationalities, relevant period, relevant variables.
- ▶ Create: eea/non-eea and sample dummies.
- ▶ Harmonize: if socio-demographic change over time, take latest value.

Mouvements de Main d'Œuvre → *wage data*.

- ▶ Filter: relevant people, contracts after 2019, longer than 30 days.
- ▶ Create: period dummies, extract the region of the contract from department codes.
- ▶ Harmonize: wages.

Shortage lists → *treatment data*.

- ▶ Scrape + clean shortage list.
- ▶ Map FAP codes (in the policy) to PCS codes (in MMO).

NB: ForCE 2025S1 data.